

Vision Transformers for End-to-End Vision- Based Quadrotor Obstacle Avoidance

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Chi Wei, Yeh · May 4, 2026

| Outline

Central research question

RESEARCH QUESTION

For high-speed depth-based quadrotor avoidance, does combining scene-wide spatial context with temporal state yield a more reliable end-to-end controller than designs with purely local spatial encoding?

[**Introduction**](#)[**Method**](#)[**Experiment**](#)[**Conclusion**](#)

| Introduction

Target task

- High-speed **forward flight** in cluttered, unknown scenes
- Input: depth image, attitude, and forward speed
- Output: commanded linear velocity in world frame

Policy view

$$(im_{\text{depth}}, q_{\text{att}}, v_{\text{fwd}}) \rightarrow v_{\text{pred}} \in \mathbb{R}^3$$

Research Scope

- Explore end-to-end perception-to-command control as an alternative to modular pipelines
- Compare **CNN / U-Net / recurrent / ViT** backbones under the same task
- Test whether **global attention + memory** helps as speed increases

Contributions

- First use of **ViT** for end-to-end quadrotor avoidance
- Five-model comparison under one imitation-learning setup
- Sim + hardware validation up to **7 m/s**

Method

Depth-to-Control via Teacher-Student Imitation

Teacher-student formulation

$$\pi_{\text{expert}}(s_t, o) = a_t$$

$$s_{t+1} = \mathcal{T}(s_t, a_t), \quad a_t \sim \pi_{\text{expert}}(s_t, o)$$

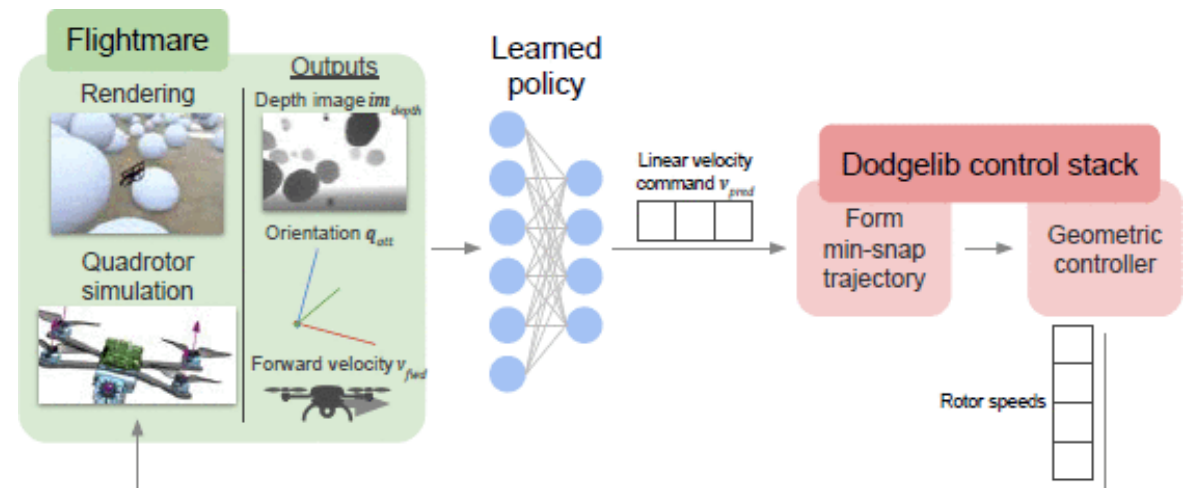
$$\pi_{\text{student}}(s_t, \text{im}_{\text{depth},t}) = a_t$$

Supervision target

$$L(\theta) = \mathbb{E}_{\tau \sim \mathcal{D}} \left[\frac{1}{T} \sum_{t=0}^{T-1} \|v_{\text{cmd}}(t) - v_{\text{pred}}(t; \theta)\|_2^2 \right]$$

- Student sees depth, not obstacle geometry
- Input size: **60 x 90**

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Shared I/O template

$$\varphi(\text{im}_{\text{depth}}) \text{ op } [q_{\text{att}}, v_{\text{fwd}}] \rightarrow v_{\text{pred}}$$

| Method

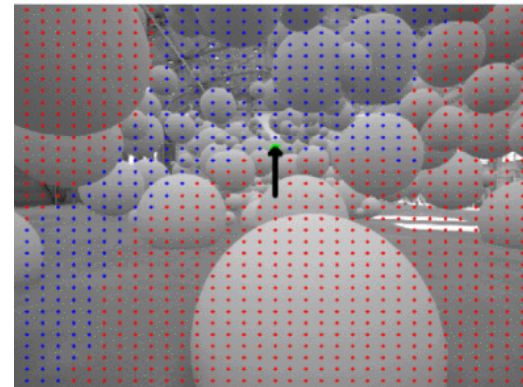
Short-Horizon Expert with Privileged Obstacle Access

Reactive expert policy

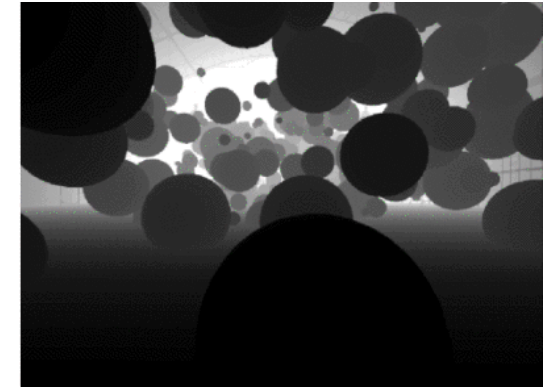
1. Build a waypoint grid ahead of the drone
2. Query collision for each candidate
3. Pick the free point nearest the center
4. Convert relative position to v_{cmd}

Privileged observation

- Obstacle positions/radii within **10 m**
- Reactive, not dynamics-feasible
- Collision trials are kept in the dataset



(a) Expert waypoint selection.



(b) Collected depth image.

588

rollouts

112k

depth
frames

27.5k

collision
frames

| Method

Spatial Inductive Bias × Temporal Memory

Controlled comparison design

encoder $\rightarrow$ fusion with q_{att} , $v_{\text{fwd}} \rightarrow$ optional LSTM $\rightarrow v_{\text{pred}}$

- Same task and supervision, mostly **3M parameters**
- Difference comes from **spatial bias** and **temporal memory**

Attribute map

Model	Spatial bias	Memory	Params
ConvNet	local	–	235k
LSTMnet	local	yes	2.95M
UNet+LSTM	multi-scale	yes	2.96M
ViT	global	–	3.10M
ViT+LSTM	global	yes	3.56M

| Method

Segformer Encoder with Recurrent Head

ViT encoder

- Segformer-inspired encoder with **2 hierarchical transformer blocks**
- Builds multi-scale spatial context across image patches
- Produces the visual latent representation before control fusion

Temporal and state conditioning

LSTM

- Multi-layer LSTM before the FC head
- Maintains temporal state across timesteps
- Encourages smoother velocity commands

State conditioning

- Quaternion q_{att} and scalar v_{fwd} are concatenated at fusion
- Keeps the policy grounded in current flight attitude and target speed

| Experiment

Evaluation Protocol and Platforms

3-7 m/s

flight speeds

90

Spheres trials / model

50

Trees trials / model

30 Hz

onboard CPU
inference

Simulation evaluation

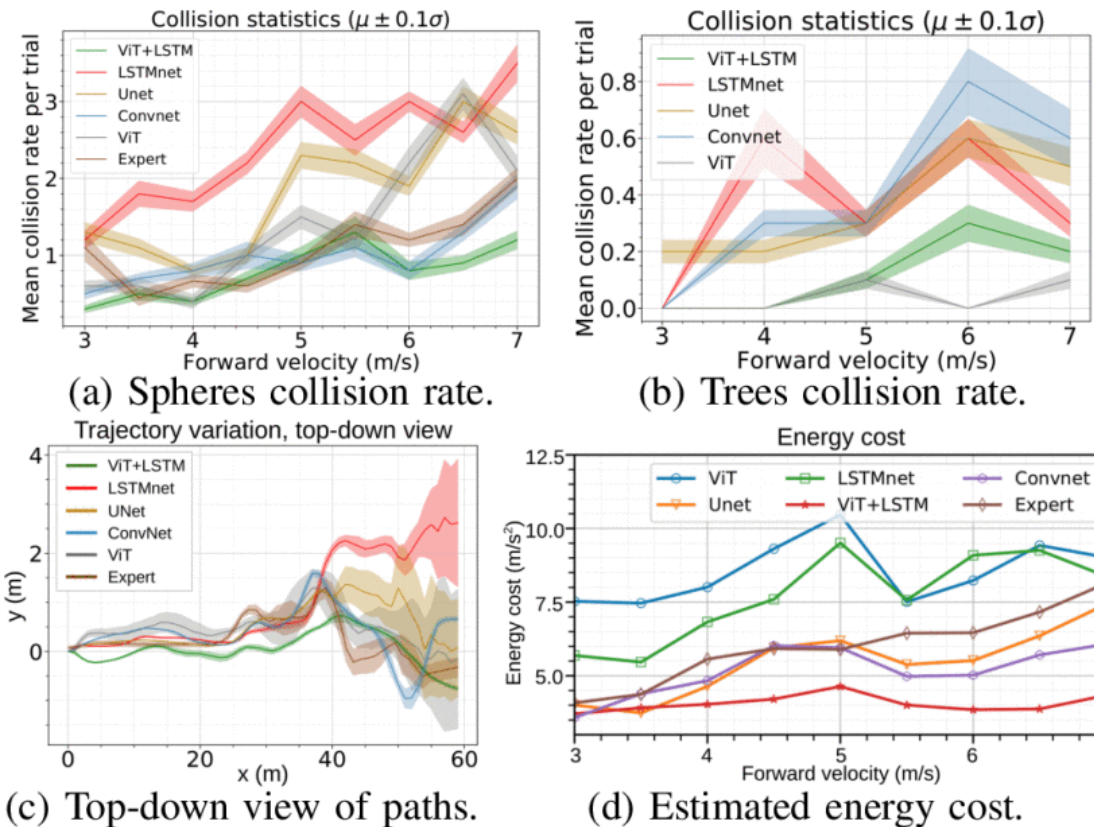
- Flightmare physics + Unity depth rendering
- **Spheres**: matched obstacle distribution for in-distribution testing
- **Trees**: unseen tree models for zero-shot generalization
- Metrics include collisions, path variation, and energy cost

Hardware platform

- Falcon250 in a motion-capture arena
- Intel RealSense D435 + Intel NUC 10 for onboard CPU inference
- Predicted velocity is tracked by the SO(3) control stack
- Real obstacles are cylinders and blocks, not floating spheres

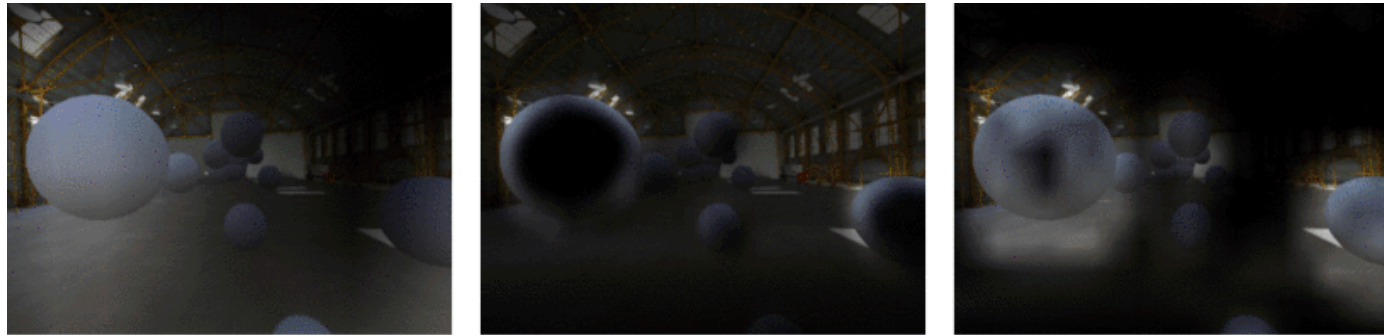
| Experiment

Backbone Comparison across Speed and Generalization

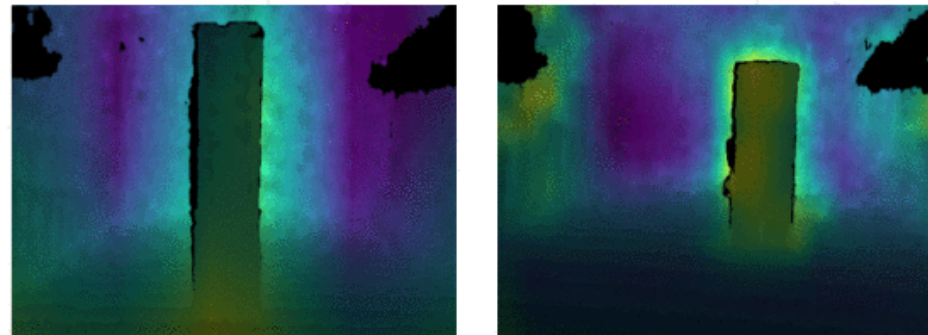


| Experiment

Activation Map Analysis across Architectures



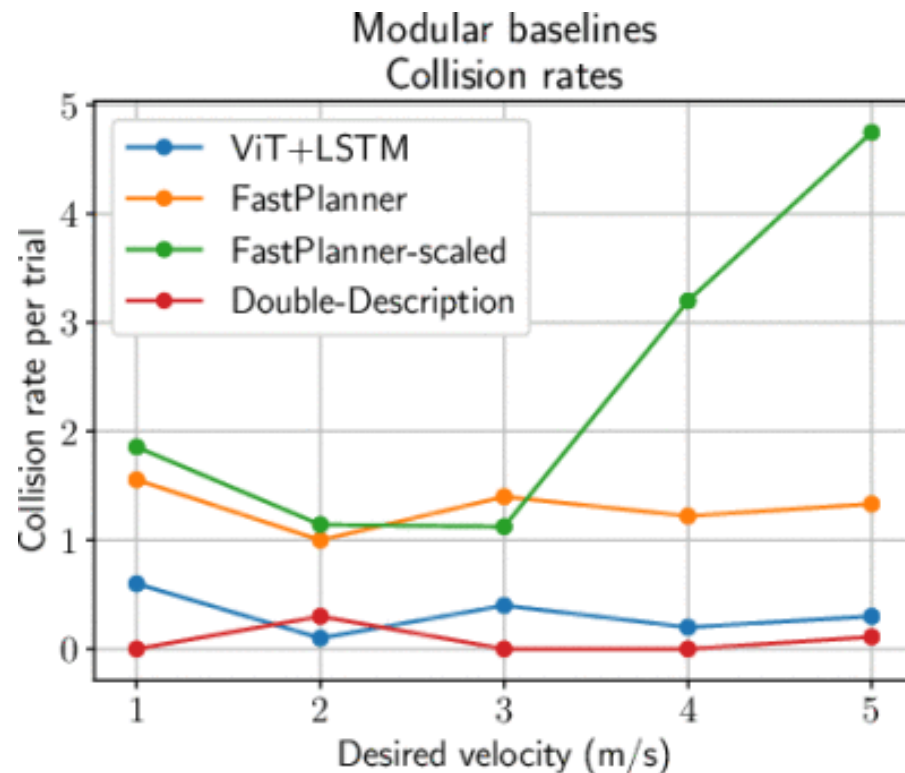
(a) Sim ConvNet. (b) Simulated UNet. (c) Simulated ViT.



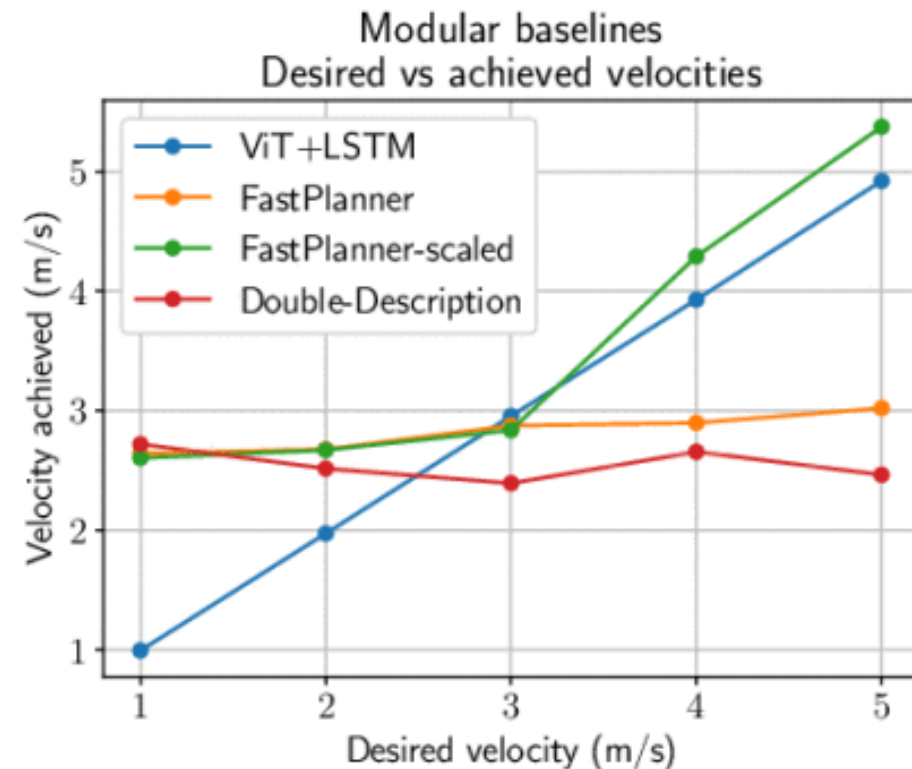
(d) Real ConvNet. (e) Real ViT.

| Experiment

Scaling against Modular Baselines



(a) Collision rates.



(b) Velocity matching.

| Experiment

Ablation against State Information

Spheres success rate (%)

Model	3-	3+	5-	5+	7-	7+
ConvNet	27	63	18	54	10	36
LSTMnet	27	45	18	9	9	9
UNet+LSTM	36	27	18	9	27	9
ViT	81	54	18	27	18	18
ViT+LSTM	54	72	45	45	9	36

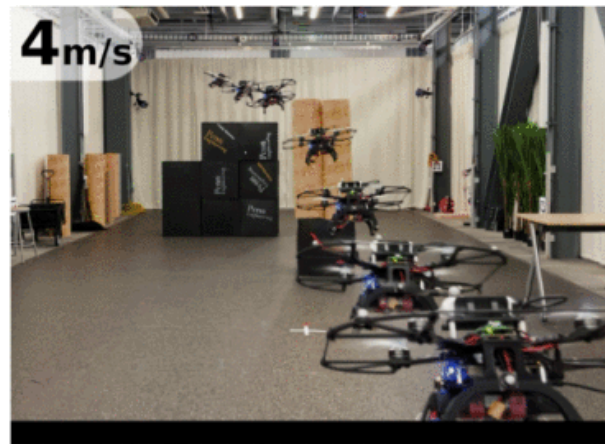
Trees success rate (%)

Model	3-	3+	5-	5+	7-	7+
ConvNet	90	100	81	63	60	42
LSTMnet	90	90	40	63	36	63
UNet+LSTM	100	72	60	63	45	54
ViT	90	100	81	81	81	80
ViT+LSTM	90	100	81	80	50	70

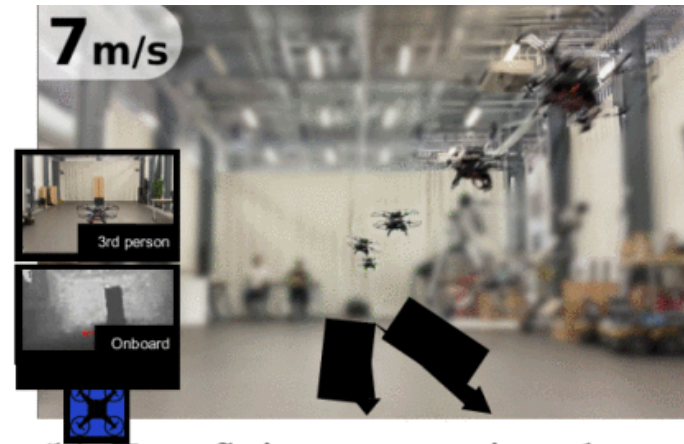
3-/5-/7-: without q_{att}, v_{fwd} ; 3+/5+/7+: with state conditioning.

| Experiment

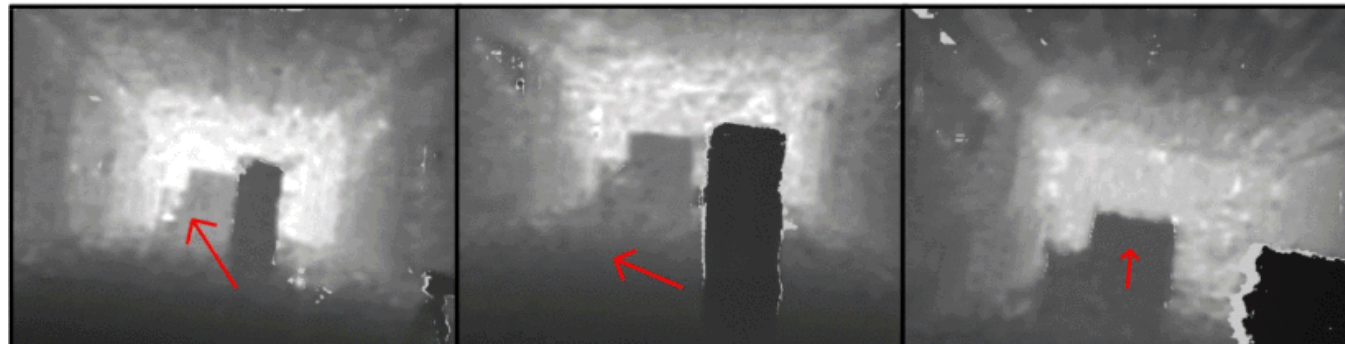
Zero-Shot Hardware Transfer



(a) Evasive maneuver.



(b) Fast-flying, oncoming drone.



| Conclusion

Answer to the central question

For this task, combining scene-wide spatial context with temporal state produced the most reliable controller, especially at high speed and under environment shift.

Empirical takeaway

ViT+LSTM is the only model that consistently surpasses the expert beyond **5 m/s** and clearly leads other students beyond **6 m/s**.

Why this matters

The paper isolates **spatial context** and **temporal state** under one imitation-learning protocol.

Scope and limitation

Evidence includes zero-shot hardware transfer, but only in a controlled motion-capture arena.